MEDIUM

PROBLEM-SOLVING AND DATA ANALYSIS

Two-variable data: Models and scatterplots

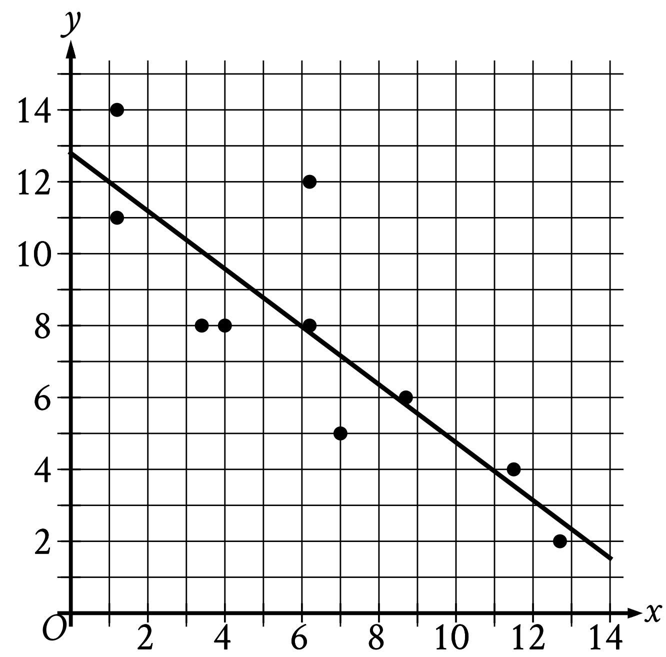
01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
    - The line of best fit slants down from left to right.
    - 2 points are touching the line of best fit.
    - 3 points are above the line of best fit.
    - 5 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 12.9)
        - (6 comma 8)
        - (12 comma 3.2)

Which of the following is closest to the slope of the line of best fit shown?

A. -2.4

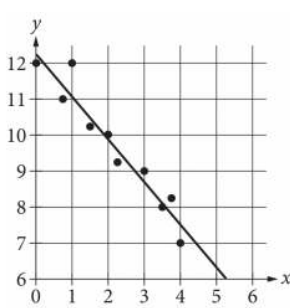
B. -0.8

C. 0.8

D. 2.4

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown. Which of the following is closest to the difference between the y-coordinate of the data point with $x = 1$ and the y-value predicted by the line of best fit at $x = 1$?



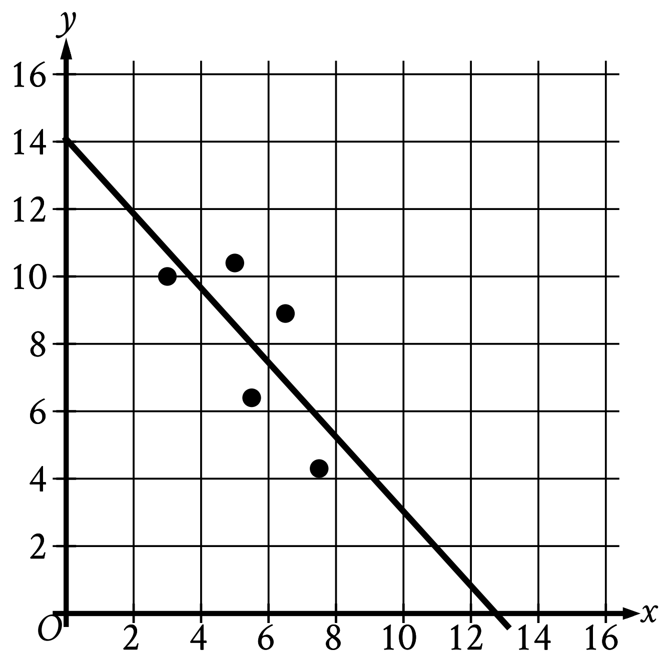
A. 1

B. 2

C. 5

D. 12

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
  - The line of best fit slants down from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 14.1)
    - (12.7 comma 0)

Which of the following is closest to the slope of this line of best fit?

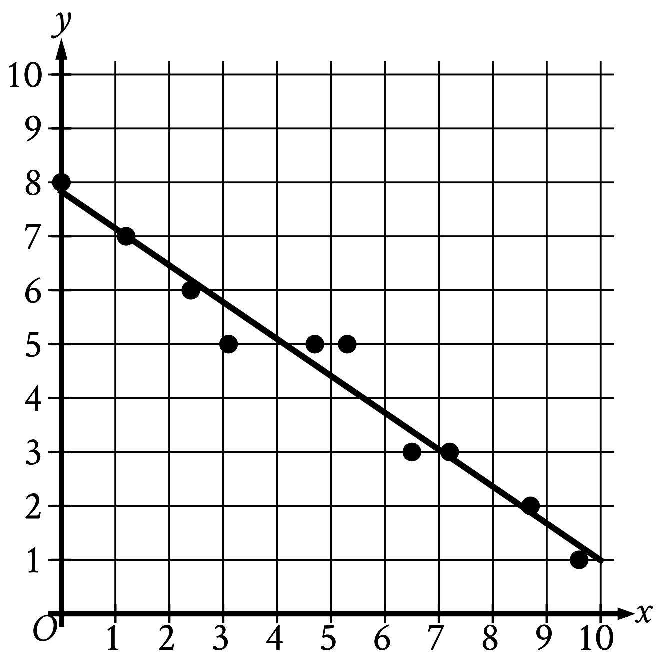
A. -3.3

B. -1.1

C. 1.1

D. 3.3

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
- The line of best fit slants down from left to right.
    - 5 points are touching the line of best fit.
    - 2 points are above the line of best fit.
    - 3 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (1 comma 7)
        - (4 comma 5)
        - (10 comma 1)

Which of the following is closest to the slope of this line of best fit?

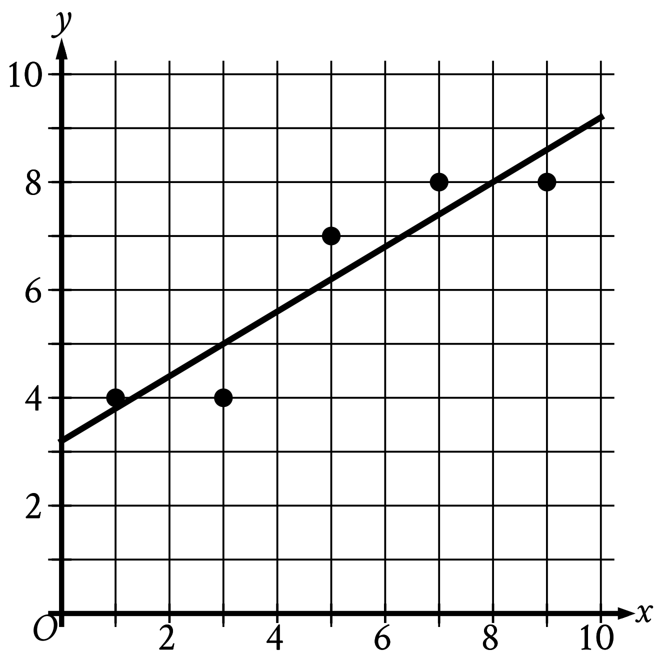
A. 7

B. 0.7

C. -0.7

D. -7

The scatterplot shows the relationship between x and y. A line of best fit is also shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
  - The line of best fit slants up from left to right.
  - The line of best fit passes through the following approximate coordinates:
    - (0 comma 3.20)
    - (1 comma 3.80)
    - (2 comma 4.40)

Which of the following is closest to the slope of this line of best fit?

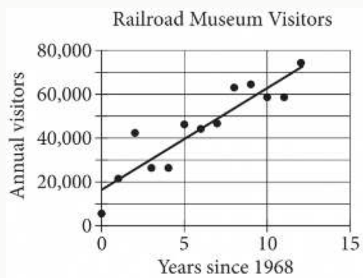
A. 0.60

B. 2.50

C. 7.80

D. 8.00

Q6 PROBLEM-SOLVING AND DATA ANALYSIS



The scatterplot above shows the number of visitors to a railroad museum in Pennsylvania each year from 1968 to 1980, where t is the number of years since 1968 and n is the number of visitors. A line of best fit is also shown. Which of the following could be an equation of the line of best fit shown?

A. $n = 16{,}090 + 4{,}680t$

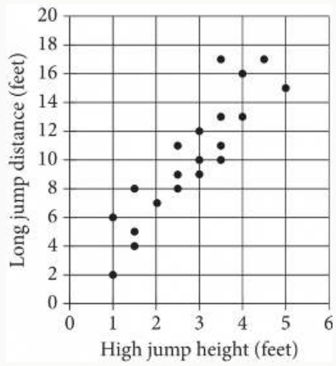
B. $n = 4{,}690 + 16{,}090t$

C. $n = 16{,}090 + 9{,}060t$

D. $n = 9{,}060 + 16{,}090t$

Q7

PROBLEM-SOLVING AND DATA ANALYSIS



Each dot in the scatterplot above represents the height x, in feet, in the high jump, and the distance y, in feet, in the long jump, made by each student in a group of twenty students. The graph of which of the following equations is a line that most closely fits the data?

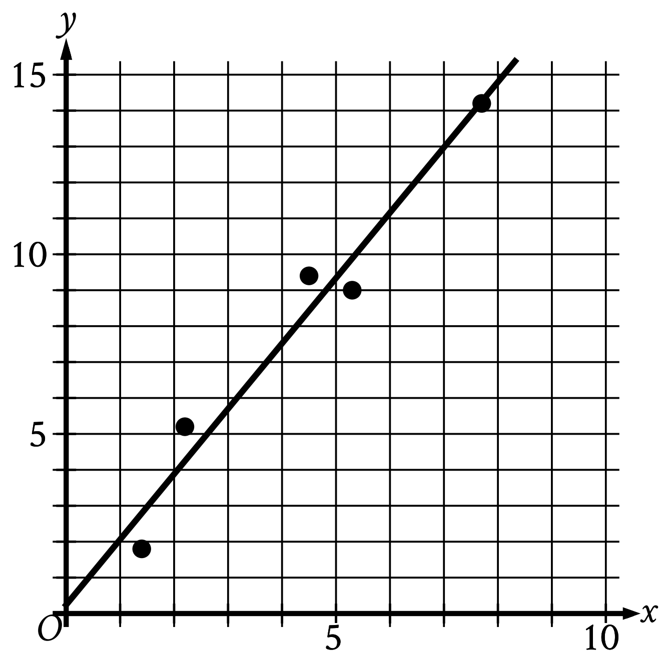
A. $y = 0.82x + 3.30$

B. $y = 0.82x - 0.82$

C. $y = 3.30x + 0.82$

D. $y = 3.30x - 3.30$

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 5 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 0.2)
        - (5 comma 9.3)

Which of the following is closest to the slope of the line of best fit shown?

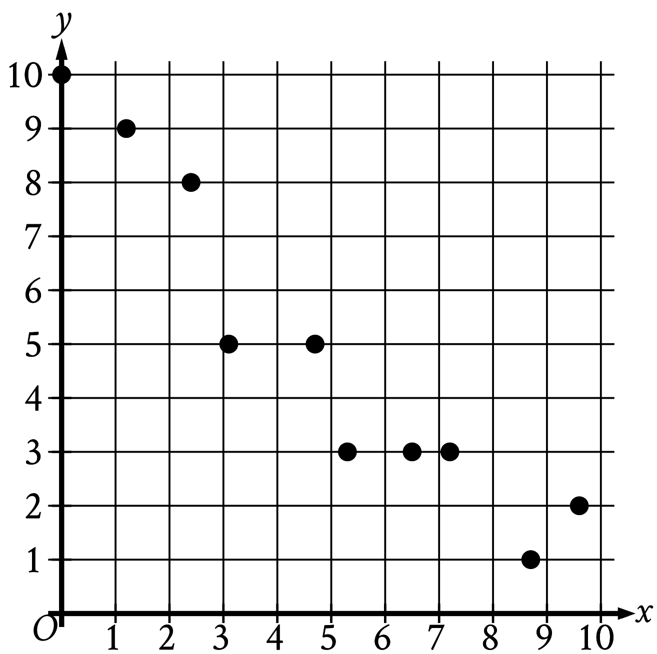
A. 0.2

B. 0.7

C. 1.8

D. 2.6

The scatterplot shows the relationship between two variables, x and y.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- The data points have the following approximate coordinates:
    - (0 comma 10)
    - (1.2 comma 9)
    - (2.3 comma 8)
    - (3.1 comma 5)
    - (4.8 comma 5)
    - (5.2 comma 3)
    - (6.5 comma 3)
    - (7.2 comma 3)
    - (9.6 comma 2)
    - (8.8 comma 1)

Which of the following equations is the most appropriate linear model for the data shown?

A. $y = 0.9 + 9.4x$

B. $y = 0.9 - 9.4x$

C. $y = 9.4 + 0.9x$

D. $y = 9.4 - 0.9x$

Q10 PROBLEM-SOLVING AND DATA ANALYSIS

Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

A. Decreasing exponential

B. Decreasing linear

C. Increasing exponential

D. Increasing linear